

DISCUSSION

SESSION 4A

DR. D. OKRENT, *Session Chairman*

To a considerable extent this session will deal with the more basic aspects of fast reactor critical experiments, the analyses thereof, and theoretical techniques, although as design aspects important to particular considerations come up, they will be dealt with as they are pertinent.

Presentation of Papers

R. AVERY—*Paper 4A/1*

The major experimental efforts in the U.S. programme for fast reactor physics centre on basic cross-section measurements, critical experiments, reactor experiments and experiments related to safety.

The increasing size of fast reactors and safety considerations particularly stimulated studies of neutron cross-sections in the resonance region. The majority of the work in the energy region up to 10 keV has exploited intense pulsed neutron sources and time-of-flight techniques to achieve excellent sensitivity. The measurements have been principally limited to several large accelerators but also include a nuclear explosive which releases several moles of neutrons in a fraction of a microsecond. The intensities available with an explosive make it possible to measure fission cross-sections in the resonance energy region, particularly those of the plutonium isotopes, with small and very active samples. The proposed new ORNL linac will lead to increased effort in this most important energy region.

Experiment in the intermediate region, between 50 keV and 1.5 MeV, has been characterized by improved use of established techniques rather than the introduction of new concepts. Representative of recent progress is the greatly improved knowledge of both elastic and inelastic neutron scattering in the intermediate region over a large portion of the periodic table.

In the energy region above 1.5 MeV, the most serious uncertainties are between 1.5 and about 3.0 MeV—the region between the discrete structure observed at intermediate energies and the lower limit of valid experimental and theoretical interpretations based on a nuclear temperature. An increased

cross-section measurement effort in this energy region is planned with a new accelerator at ANL.

ZPR-3 (ANL, Idaho) and ZPR-6 and 9 (ANL, Illinois) are carrying out critical experiments in support of the U.S. fast power breeder reactor programme. The Atomics International facility, which allows the study of fast zones surrounded by thermal driving regions, has also been used in the programme, particularly for the measurement of the Doppler effect in the past several years. All of these facilities and the planned Argonne ZPPR, to be used for large scale plutonium assemblies, are highly flexible, horizontally loaded, split table machines.

Recent assemblies have concentrated, in addition to mockups of specific reactors, on three general problems: the Doppler effect, the sodium void effect, and the systematics and validity of zoned studies.

Doppler measurements by the exchange of hot and cold samples have been made in the past several years in the AI facility and in ZPR-3 and 6 at Argonne. The Doppler measurement programme at AI includes measurements of U^{235} and U^{238} in various spectra at different sample temperatures.

Two sequences of Doppler effect measurements have been made in ZPR-3, the first in U^{235} fuelled cores, and the second in plutonium fuelled cores. In the plutonium fuelled case the experiments were done in a reactor whose central zone corresponded to the composition of a large uranium-plutonium (7:1) monocarbide, and to a graphite softened version of the same. The Doppler samples studied consisted of 0.50 in. diameter pellets of oxide sintered to 0.70 theoretical density, and stacked to a height of 6 in. Enriched Pu^{239} , mixed U^{238} and Pu^{239} (7:1), depleted uranium, and enriched uranium samples were used.

The Doppler measurements have since been continued and extended in U^{235} fuelled assemblies in ZPR-6. The phenomena studied related to the effect of an isolation zone around the sample, the effect of restraint on expansion, size effect of sample, effect of having diluent interspersed with sample and effect of void region surrounding the sample. The measurements were made with various sample fuel enrichments as a function of sample temperature.

The sodium void effect on reactivity for large fast reactors is important from safety considerations. It has been measured in the Fermi reactor, on assemblies constructed on ZPR-3, and more thoroughly on two large carbide assemblies on ZPR-6. The first assembly studied on ZPR-6 had a 610 litre uranium carbide core with a length-to-diameter ratio of approximately unity. The second assembly had a flat pancake core with a volume of about 950 litres and a length-to-diameter ratio of about 1:3. In both cores, a thorough mapping of the sodium void coefficient as a function of location has been accomplished.

Experiments to study the heterogeneity effect on the sodium coefficient were done in a central region of the 950 litre core assembly. Included were:

different materials placed next to the sodium; different widths of sodium columns; partial voids in a 1 in. wide sodium column; and discontinuous or staggered void in the normal drawer loading. As much as about 20% variation between the highest and lowest value for the sodium coefficient was obtained with the various configurations. It would appear that for larger dilute cores, where the spectrum is more degraded, it is even more important to take into account the heterogeneity effects.

A 1100 litre core volume, four radial region reactor has been studied on ZPR-6 to determine the extent to which physics parameters of a large full scale reactor may be predicted on the basis of measurements in a zoned system. This assembly was composed of central, buffer, driver and reflector radial regions.

The central region composition was chosen to be identical to that for the 2600 litre core about to be studied on ZPR-6. The composition of the buffer and driver regions were chosen so that there would be a good calculated matching of the real and adjoint fluxes between the zoned core and the full scale 2600 litre system.

Central core spectrum using the proton recoil spectrometer developed by Bennett, fission ratio measurements through the central zone, reactivity worth measurements of different material including core materials in the central, buffer and driver regions, are some of the measurements made at the centre of the zoned assembly and are on the full scale assembly. In addition, comparative Doppler coefficient measurements of U^{235} and U^{238} and sodium void coefficient studies are to be made in the two assemblies. The loading of the full scale system is to begin shortly.

The SEFOR project is intended to obtain physics and safety information on an operating fast ceramic fuelled reactor. Particular emphasis is to be paid to the measurement and analysis of the Doppler effect under steady state and transient conditions.

The 20 MW(th) SEFOR reactor is sodium cooled and utilizes uranium-plutonium oxide fuel containing approximately 20% plutonium. The core is in the shape of a cylinder, approximately 34 in. in diameter and 34 in. high. Fuel pellet diameter in an individual pin is approximately 0.9 in., contained in a cladding of 0.040 in. stainless steel. The diameter of the fuel rods is made large so that at the relatively low SEFOR power density the temperature distribution within the rod is very close to that expected within the fuel of a large fast ceramic breeder reactor. Similarly, the core composition is chosen so that the neutron spectrum and the Doppler coefficient are quite close to that expected in a large fast breeder power reactor. This is accomplished primarily by utilizing a high volume fraction of the core for fuel. The planned test programme for SEFOR has been divided into four parts. After the usual initial critical measurements, a series of steady state tests will take place. These tests involve the measurement of the change of reactivity between two

different reactor operating conditions. For example, one test involves operation of the reactor at 10 MW(th) at half of full flow rate and then operation of the reactor at 20 MW(th) under full flow conditions. Since the sodium temperature remains essentially unchanged between these two conditions, the reactivity differences can be attributed—with small corrections—to the change in fuel temperature.

The second set of measurements are oscillator tests involving the oscillation of a special control element. At low frequencies—several cycles per minute or less—the Doppler feedback depends upon the Doppler coefficient divided by the fuel conductivity. These measurements will provide information similar to that obtained from the steady state tests, and thus provide a check on these tests. At high frequencies—from 10 c/m to 1 c/s—the feedback response depends upon the Doppler coefficient divided by the heat capacity of the fuel. This is the same quantity which enters into the transient behaviour of the reactor.

The transient tests are divided into two parts: sub-prompt critical tests and prompt critical tests. In the sub-prompt critical tests, reactivity insertions below β are involved. Results of this test again involve the Doppler coefficient divided by the heat capacity, and the analysis is made in terms of the delayed neutron parameters.

Following completion of the sub-prompt critical tests, a series of above-prompt critical excursion tests will be performed. In this case, the analysis involves the prompt neutron lifetime and not the parameters of the individual delayed neutron precursors.

The SEFOR reactor is now under construction. The reactor containment has been completed, and internal structural work is in progress. The plant components, such as heat exchangers, pumps, reactor vessel, and so on, are being fabricated. An associated preoperational research and development programme involving fuel testing and tests on key elements of the plant is expected to be completed in 1966. ZPR-3 critical experiments on SEFOR have recently been completed. Initial dry critical testing of SEFOR is expected to take place at the end of 1967.

Investigations into the phenomena and mechanisms of core meltdown accidents form the nucleus of the experimental fast reactor safety programme. Analysis of a meltdown incident requires knowledge of the causes of fuel failure and movement, the modes of failure and material movement, rates of movement, and degrees of coherence in movement. Many complex and little understood phenomena are involved, including fuel slumping, cladding dissolution by molten fuel, transient stresses, fission gas effects, interactions between pins in clusters, and coolant boiling. For reactor designs with appreciable sodium void reactivity effects, coolant expulsion must be understood for its effect on reactivity as well as on fuel movement.

The experimental programme is centred on the transient reactor tests

facility—TREAT—which is being used to identify phenomena of interest, study their significance and inter-relationship under in-pile conditions, and provide tests of theoretical predictions. Out-of-pile experimentation is being relied upon for model tests, basic data, and extension of in-pile investigations. The basic philosophy is to separate the various problems, such as pre-failure fuel motions, and fission gas effects, and study them as systematically as possible. Out-of-pile studies are being made on the physical and chemical interactions resulting from sudden immersion in sodium of molten fuel materials. A broad range of heat transfer experiments is in preparation, including studies on transient expulsion from heated sections, critical flow phenomena under accident conditions, superheat, and transient heat transfer from small spheres moving rapidly in liquid and boiling sodium.

Work is being initiated on filling in the most serious gap in the equation-of-state data, guided by theoretical evaluations of the available information. TREAT experiments have been made on the following fuel types: metallic fuels, including EBR-II Mark I, Fermi Core A, and EBR-II Th-U; and also on ceramic fuels, including UO_2 , mixed oxide, and UC. Tests are planned and will continue to be planned for all fuel elements of potential interest to the fast breeder programme.

S. YIFTAH—*Paper 4A/2*

A large Doppler coefficient is essential to ensure stability and excursion limiting capacity for large fast reactors.

Calculations of the Doppler effect is complicated, since it depends on knowledge of the principles and of much of the data of basic nuclear physics (e.g. resonance and fission phenomena), and also involves the main analytical techniques of the reactor physicist (i.e. multi-group diffusion, transport and Monte Carlo theories). Most calculations are performed with the help of computer codes, and therefore applied mathematicians and computer programmers are called upon.

In the near future the Doppler effect will be measured in many places, notably in the SEFOR project, which is being built mainly for this purpose. A comparison of theoretical and calculated results with experimental results will then be possible.

Work on the Doppler effect started in Israel in 1962. Since 1964 research has been partly supported by the Karlsruhe-Euratom Association on Fast Reactors. Some aspects and results of this work have been summarized and reported in the papers listed in the references (p. 430). The main effort was concentrated in four groups of problems—computation methods; temperature-dependent effective cross-sections; parametric-type calculation of the Doppler effect on various systems; and special effects such as resonance interaction (fuel, coolant and structural material resonances), heterogeneity, multi-level formulae for fissile nuclides.

Three different methods of calculating the Doppler effect were developed and coded for the Philco-2000 computer. These are based on the successive- k , perturbation and Monte Carlo techniques.

All calculations of the Doppler effect depend largely on nuclear parameters and basic resonant data. Lack of sufficient experimental information and the need to extrapolate low energy data to high energies with debatable or not fully justified methods makes it difficult to choose a suitable nuclear physics input for the calculations. There is also the constant problem of re-evaluation to include the most recent experimental and theoretical information.

The uncertainties in the physical data are themselves difficult to assess, as is the effect of these uncertainties on the calculations.

The systems calculated to date include spherical 800–2500 litre reactors fuelled with plutonium and uranium in the metallic, oxide and carbide forms. Different isotopic compositions of the plutonium have been investigated. The above core volumes are in the range of the prototype fast reactors (250 MW(e)) to be built in the United Kingdom, the United States, Germany and France. Also, the Russian reactor BN-350 dual purpose power/desalination fast oxide breeder being built near Shevchenko and having a core volume of 1870 litres is in the same core volume range. The core volumes studies are bigger than those of experimental fast reactors at present operating in the U.K. the U.S.A. and the U.S.S.R., and smaller than the contemplated 1000 MW(e) 5000–8000 litre cores.

The specification of the systems, together with the range of changes of some of the parameters, is given in Table 1 of Paper 4A/2 which shows the plutonium isotopic composition and the metallic blanket used in this calculation.

The static parameters and breeding potential of these systems were studied a few years ago by Dr Okrent and myself at Argonne. Later, some of the dynamic parameters of the same system were calculated and published in an Argonne report.

The calculation results using the successive- k method are summarized in Table 2 of the paper. Table 3 of the paper gives a comparison of the successive- k and perturbation methods for plutonium A—pure plutonium 239—systems.

The main conclusions to be drawn from these results can be summarized in terms of three effects—the volume, the fuel chemical composition and the plutonium isotopic composition effects.

Figure 1 of Paper 4A/2 shows the dependence of the Doppler effect on core volume for various chemical compositions of fuel. It shows that the oxide case has the highest Doppler influence, followed by carbide and metal, and that the Doppler effect goes up in all cases. The slope of the oxide case is steepest, showing that for oxide the volume effect is the strongest.

Figure 2 of the paper shows similar curves for the carbide case. It shows the plutonium isotopic composition effect, indicating that the plutonium C,

which is the richest in higher isotopes, has a higher Doppler effect than either plutonium B or pure plutonium 239.

The Doppler effect in the 800–2500 litre spherical cores described in the study is of the order of $10^{-5} \Delta k/k/^\circ\text{C}$ at 900°K and is sensitive to core volume. The sensitivity is greatest in oxide fuel. The larger the core volume the greater is the absolute magnitude of the effect.

In reactors of equal core volumes the Doppler effect in volumes above 800 litres is smallest with metallic fuel, and above 1400 litres is greatest with oxide fuels. The curve shows that for volumes smaller than 1400 litres the Doppler effect of the carbides is slightly higher than that of the oxides.

The presence of higher plutonium isotopes— Pu^{240} , Pu^{241} and Pu^{242} —in general increases the absolute value of the effect. Pu^{239} and Pu^{241} have a positive contribution, while U^{238} , Pu^{240} and Pu^{242} contribute negatively. The overall effect depends on the relative contribution of each of the isotopes present; it changes with Y^* , the ratio between the amounts of $\text{U}^{238} + \text{Pu}^{240} + \text{Pu}^{242}$ and $\text{Pu}^{239} + \text{Pu}^{241}$ as seen in Table 2 of the paper. It should be noted that the influence of the positive effect of Pu^{239} and Pu^{241} decreases with the increase of the content of Pu^{241} since the latter is a better fissile material than Pu^{239} (i.e. greater ν and smaller α) and thus the total mass of fissile isotopes for criticality is smaller if the plutonium contains greater amounts of high isotopes. For a given core volume a decrease in the critical mass will imply an increase in the total mass of non-fissile material, contributing negatively to the overall effect. Altogether the effect of differences in the plutonium isotopic composition on the total Doppler coefficient may be as much as 30–40%. This is a greater effect than one would expect.

It should be mentioned that a Monte Carlo technique was used to study the effect of the interaction between the resonances of fertile and fissile isotopes and the heterogeneity of fast reactor cores. The conclusions of these studies indicate that both these effects seem to be negligible.

The comparison of the successive- k and perturbation methods of computation shows that for large cores and for oxide and carbide fuels simple perturbation calculation is not adequate and that perturbed fluxes should also be used in the calculation.

N. V. KRASNOYAROV—*Paper 4A/3*
(Spoke in Russian)

Dr. Krasnoyarov summarized the main points of his paper.

H. H. HUMMEL—*Paper 4A/4*

The field of fast reactor physics theory in which there has been the greatest activity recently in the United States has been that of the reactivity coefficients most important for large reactors, namely the Doppler and coolant void

coefficient. There has been considerable study of fertile-fissile resonance overlap effects following the discovery of the importance of this phenomenon for the calculation of the fissile material Doppler effect by Codd and Collins.

It has been found by Nicholson and Fischer at APDA and by Hutchins at GE that it is possible to account for the overlap effects with sufficient accuracy in large, dilute reactors by using effective cross-sections for individual isotopes calculated by ignoring the presence of other isotopes. This involves the use of factors to correct group fluxes for resonant depressions which differ for different isotopes, while properly these factors should all be the same. This procedure involves an approximation in evaluation of the flux correction factors which Hwang at Argonne has preferred not to make as he has felt that it might involve some error at higher fissile material content. Whether this area would actually be important in any case of practical interest has not yet been determined. The Nicholson procedure is obviously much more convenient since it permits tabulation of cross-sections for each isotope without reference to the relative concentrations of the various isotopes present in a mixture, while Hwang's procedure requires such reference. Because of this convenience and pending further checks of its accuracy, Nicholson's procedure has been incorporated in the Argonne cross-section generating code MC².

Modest errors in the evaluation of fertile-fissile overlap effects are not too important at present because of a lack of knowledge of resonance parameters for fissile isotopes over most of the energy range of interest. This makes it necessary to rely on an assumed statistical distribution of resonance parameters to carry out calculations of the fissile material Doppler effect. Work by Hwang has shown that below about 1 keV level densities are not sufficiently high to make a statistical treatment very satisfactory, while resolved parameters for Pu²³⁹ extend only up to 300 eV. A similar conclusion was reached in studies by Kelber and Kier. They used a procedure similar to Brissenden's random selection of resonance parameters and level spacings repeated many times over, and found a broad dispersion in calculated fissile material Doppler effect.

There is at present a discrepancy between the calculated and measured Pu²³⁹ Doppler effect that has not been explained. The measured value is about zero, while calculations indicate a small positive effect of about 12% of the negative U²³⁸ one. It has been found recently that there are very broad resonances associated with $J = 0$ states in Pu²³⁹ and narrow ones with $J = 1$. This has led to some reduction in the calculated Pu²³⁹ Doppler effect, but a discrepancy still exists.

The use of this assumption has led to some reduction in the Doppler effect at present calculated. The reduction is 50% over that obtained previously. It is not believed that the above statistical uncertainty is large enough to explain the discrepancy, or that it can be accounted for by reasonable alteration of resonance parameters—perhaps a controversial statement, but it is

based on the fact that a comparison between the statistical assumption and the resolution resonance calculation up to 300 eV shows some reduction of the plutonium Doppler effect. It was felt that it was not reasonable to find out what the parameters are at higher levels. A little while ago it was found that by making certain specific assumptions about what the resonance parameters are, any Doppler effect required can be obtained.

While the theory of the fissile material Doppler effect is not reliable, it is reasonably satisfactory for the fertile material Doppler effect.

Critical experiments have shown that as the spectrum hardens, the plutonium Doppler effect becomes positive. If a little boron is added round the sample, the agreement between theory and experiment becomes much better. Any choice of resonance parameters to be taken as satisfactory will have to explain this, and I am not sure that we have anything so far which does this.

Turning to the question of the plutonium Doppler effect, it seems that the experimental value is definitely larger than the calculated one. It was 38% larger for a typical carbide fuelled reactor spectrum. This must be due to too small a calculated low energy flux. It does not seem to me that we shall have such a large error in the calculation method of the resonance parameters. The exact source of this error in the calculation spectrum is not too clear yet. It is probably a combination of U^{238} and incorrect inelastic scattering cross-section. In a spectrum over-moderated from a fast power reactor spectrum, agreement was closer. Since in the latter case the spectrum is mainly determined by elastic moderation and in the former inelastic scattering was also important, this may indicate that incorrect inelastic scattering cross-sections are causing the error in spectrum. I feel that we have to work hard to get very much out of fission and capture cross-sections. Perhaps 10% more is the maximum as far as I can see; it may be that others will disagree with this. Perhaps 10% could be obtained from the fission capture and another 10% out of U^{238} scattering.

Hwang at Argonne has investigated the accuracy of the narrow resonance approximation for resonance integral and Doppler effects, and finds that the error in Doppler effect is less than 10%. He has developed an improved analytical calculation which corrects errors in the narrow resonance approximation. The analysis of critical experiment has revealed some other indications that the calculated spectrum may be too hard, because the calculated boron reaction rate is a little too small and the calculated prompt lifetimes are all too short. There seems to be some correlation there. It does not indicate anything very big. It may be 10–15%, which might be reasonable.

In the field of reactor concepts, we in the United States have been very much concerned about the boiling effect, perhaps more so than other people. One concept that has been pursued a little lately is the idea of using a zoned core with the composition of the central part of the core selected to make the sodium void effect more negative in the region where it is normally most

positive. The most effective arrangement is one with U^{233} -Th fuel in the central part of the core and Pu-U in the outer part. Recent results have indicated that high internal breeding together with a small sodium void effect can be achieved with such a core. However, the economics of having two fuel cycles in a single core have not yet been evaluated.

An interesting feature of the Fast Reactor Conference at Argonne in October 1965 was the presentation of the results of comparison calculations

TABLE D1. SPECTRAL AND CAPTURE δk FOR 50% SODIUM REMOVAL

Organization	Case 10 oxide: $U^{238}/Pu^{239} = 5$			
	El.	Inel.	Capt.	Total
ANL, 300°K	1.01	0.64	0.12	1.77
ANL, 1500°K	1.14	0.65	0.12	1.91
AI, 1500°K	0.61	0.46	0.13	1.20
ANL, AI σ 's	0.49	0.49	0.12	1.10
APDA, 1500°K	1.17	0.54	0.16	1.87
ANL, APDA σ 's	1.15	0.64	0.11	1.90
GE, 1500°K	0.75	0.64	0.09	1.48
ANL, GE σ 's	0.72	0.61	0.11	1.44
UK, 300°K		1.84	0.11	1.95
ANL, UK σ 's	1.39	0.62	0.11	2.12
Karlsruhe, 900°K		1.89	0.13	2.02
ANL, Karlsruhe σ 's	1.26	0.68	0.11	2.05

performed by various organizations in the U.S.A. and Europe for representative fast reactor compositions. One of the points calculated was the sodium void effect for a number of different compositions. An attempt was made to understand the discrepancies, which has not been completely successful. It was found that for discrepancies at low energy ($< 100\text{keV}$) Pu^{239} fission and capture cross-section was a major factor, with low energy U^{238} capture cross-sections being important in some cases. Discrepancies in choice of Pu^{240} and fission product cross-sections played a smaller role. It appeared that differences in choices of sodium inelastic scattering cross-sections may have been important, although this has not been definitely determined. Other studies have also indicated the importance of discrepancies in the low energy Pu^{239} fission and capture cross-sections for coolant void effect and critical size calculations. Table 1 shows some of the worst discrepancies brought out by the various organizations in regard to the sodium void effect. After differences in the Pu^{239} cross-sections, the next important source of error are the U^{238} capture data, as is shown by the results from AI. Tables D1 and D2 also show the recalculation by ANL of the results from the various

organizations, using their data except for the Sodium cross-sections, which were held constant. The French used the Russian cross-sections, but they used their own calculation of the void coefficients rather than the Russian ones. Here the agreement is fairly good except for scatter, which suggests that there might be some discrepancy between the Russians and ourselves. GE was another with a fairly large discrepancy initially, but it has been satisfactorily explained. It was a combination of Pu^{239} fission cross-section and sodium. GE used a sodium elastic cross-section which was greater than ours. In the case of the U.K., there are many discrepancies which I have not explained. In the case of Karlsruhe, the agreement seems to be fairly good.

Table D2 is a more realistic case. It is more in the range of reactor com-

TABLE D2. SPECTRAL AND CAPTURE δk FOR 50% SODIUM REMOVAL

Organization	Case 8 oxide: $\text{U}^{238}/\text{Pu}^{239} = 9$			
	El.	Inel.	Capt.	Total
ANL, 300°K	2.11	1.05	0.22	3.38
ANL, 1500°K	2.45	1.10	0.21	3.76
AI, 1500°K	1.74	0.71	0.22	2.67
ANL, AI σ 's	1.75	0.85	0.22	2.82
France, 900°K	2.31	0.89	0.15	3.35
ANL, French σ 's	2.53	1.20	0.18	3.91
GE, 1500°K	1.63	1.04	0.15	2.82
ANL, GE σ 's	1.63	1.03	0.20	2.86
U.K., 300°K	3.20		0.20	3.40
ANL, U.K. σ 's	2.57	1.09	0.21	3.87
Karlsruhe, 900°K	3.37		0.27	3.64
ANL, Karlsruhe σ 's	2.51	1.13	0.20	3.84

positions that people are interested in building. The discrepancies are not as great and the only unexplained ones do not amount to very much. I feel that this situation is not as desperate as has been made out. I believe that we are on the road to understanding it and doing something about it. The worst cases seem to arise from the plutonium fission cross-section. None of the calculations shown has used the more recent Pu^{239} fission cross-sections, which are lower than any of those used here, and seem to have more positive sodium void effect. There is some uncertainty about it in a number of energy regions, and certainly below about 500 eV. I think that the Argonne assumptions may have been too low. The resonance parameters go up to about 300 eV. Above that there is a range where there is a variation in the cross-section and the situation is not too good.

R. D. SMITH—*Paper 4A/5*

There is still considerable difficulty in obtaining the accuracy required by designers. For example, a designer will want to know the enrichment to which he has to make his fuel within about 1%. He will want to know the worth of his control rods, the reaction rates, the heat productions and the temperature coefficients, first to an accuracy which will enable him to make sure that the reactor is safe, and then to a greater degree of accuracy as he tightens up the design.

The data for this are inadequate, such as the plutonium fission cross-section data, as the author of paper 4A/4 has pointed out. There are still discrepancies of about 30% in the fission cross-section in the 10–100 keV range. It is surprising that after so many years of atomic energy power stations all over the world, we still do not know the plutonium fission cross-section!

These discrepancies draw attention to a question which will be of growing importance in the years to come, that is, how far the data measurers can improve the accuracy of their measurements. Will they achieve sufficient accuracy for all the needs of the designers? If not, we may have to adopt a system of data adjustment based on integral measurements, and we are looking into the development of such a system in the U.K.

Basically, the data we use are from the United Kingdom Data File. We have extracted from this a thirty-two group set, which we call the FD2 set, and which has half lethargy groups covering a range down to 0.4 eV. This replaces the older FDI set.

As well as the data limitations, there are calculation limitations. The size of available computers is inadequate: 3-D calculations cannot be done in any large number of groups, even 2-D calculations cannot be done in a sufficient number of mesh points. This gives rise to some difficulties in calculating a reactor, particularly in the geometrical complexity of an engineering reactor.

Heterogeneity also causes a great deal of trouble and the methods of dealing with it are not fully developed. It seems from the work done at Winfrith that in a power reactor where the pins are of small diameter the heterogeneity effects are perhaps not too serious, but they may have quite big effects in zero power reactor assemblies such as Zebra core 6. The U^{239} fission cross-section varied across the unit cell by about 10% and the U^{238} capture cross-section varied by about 25%. This makes interpretation and accurate comparison with theory difficult.

Calculation of control rod worth is still difficult. We are developing Monte Carlo methods to do this. At the moment we have semi-empirical methods which are in some cases, I think, a little better than guesswork. Blanket calculations are also difficult, particularly if moderating material is present.

Table 5 in Paper 4A/5 deals with the question of enrichment. We have calculations of various systems which have been built in Zebra. Assembly

No. 1 is a small uranium assembly; assembly No. 2 is a big uranium assembly; assembly No. 3 is a plutonium assembly; assembly No. 6A is another plutonium assembly which we have just finished. We have managed to calculate criticality fairly well for all these systems.

Figure 3 of the paper shows some reaction rates along the axis of the Zebra assemblies. In general, these give very good agreement. The experimental points lie on the calculated curve with some exceptions at the region boundaries, where the points tend to lie off the curve. Bearing in mind the method of calculation, this is not unexpected. It points to the fact that we need to develop better methods of treating effects in regions where there are spectrum changes.

Figures 6 and 7 of Paper 4A/5 show the Zebra blanket in which was a graphite zone (which acted as the core reflector); outside this was the breeder zone, and then some more graphite and steel. The agreement between the calculation and the experiment was worse than in the previous case of a simple natural uranium reflector. There are considerable difficulties, both in the calculation and in the experimental measurement, when low energy neutrons are present. The core reflector, however, improves the radial form fast and concentrates the breeding into a smaller amount of uranium.

Table 6 of the paper gives some comparisons of experimental fission ratios and theoretical fission ratios. By using the data set very good agreement between experiment and calculation is generally obtained. In particular, just as good agreement is obtained for plutonium fuel systems as for uranium fuel systems. There are, however, apparently too few fast neutrons in these calculations because the 240 and 238 fission rates are invariably calculated in error. We suspect that this is due both to the 239 and the 235 fission data being in error, and also to the 238 inelastic scatter cross-section. As a result of these various errors, we still feel that it is necessary to build mockups, and we are at present building one of the PFR for Zebra. The mockup has about 300 kg of Pu^{239} and 450 kg of U^{235} , but it appears that our criticality calculations are not exactly correct. We expected it to go critical at a radius of about 145 cm. The latest extrapolation shows 156 cm, which corresponds to an error of the order of about 1% in reactivity. It therefore appears that the PFR will be slightly less reactive than was calculated. The error of 1% in reactivity is perhaps not too bad for such a large system.

Discussion

J. B. NIMS (A.P.D.A.): I should like to ask Dr. Smith, Paper 4A/5, for his opinion on the relative merits of measuring the Doppler coefficient by means of the hot central loop compared with the small sample oscillation technique.

R. D. SMITH—*Paper 4A/5*: We went to a great deal of trouble to build a large loop in Zebra, which at least implied that we believed that this was

the right way to do it. These experiments must be viewed against the background of the circumstances. The circumstances were that we wanted an assurance that the Doppler coefficient would be acceptable in the prototype fast reactor. The interpretation of small Doppler experiments leads to a great deal of argument about hot and cold zones and the interaction of resonance in and outside the sample.

We wanted to do the nearest thing to heating the whole reactor. The loop heated was the biggest that was practicable. Having once done that experiment and compared the result with small sample experiments, it seems in general that the small sample experiments give very similar results to the big loop. I think that the small sample experiments now will give a great deal of the additional information. In other words, these experiments are complementary, and they are not alternative methods of measuring the same things.

J. B. NIMS: Perhaps you can give us a little more information. Either technique has set a whole problem in interpreting the measurements, particularly for the fissile isotopes. Have you any feeling about in which case the error is more serious?

J. L. ROWLANDS—*Paper 4A/5*: The interpretation of the Doppler effect in a fissile sample is more difficult than the fissile effect in a zone, because the stronger shielding in a sample has two effects. First, there is the effect by virtue of the stronger shielding, and, secondly, the effect tends to be shifted to a higher energy region. There is evidence that the Doppler effect at higher energies is more positive than that at lower energies.

CHAIRMAN: Did you say that with the heated zone it is more difficult to make the measurement with the fissile material?

J. L. ROWLANDS: No. It is more difficult to interpret the result for a small sample or to relate the result for a small sample to the effect for a reactor, because the resonances in a small sample are much more strongly shielded than they are in the usual reactor environment, and this has an additional effect of shifting the Doppler effect to a higher energy region than exists in the reactor. There is some evidence that the Pu^{239} Doppler effect, for example, is positive in the keV region but possibly negative in the hundreds of eV region. If a measurement is made with a small Pu^{239} sample a positive effect might be obtained, but if one makes a measurement with a zone, one might get a negative effect. The problem is extrapolating from the sample result to the reactor result.

CHAIRMAN: Can one make a measurement in a large heated zone in which one heats up the plutonium?

J. L. ROWLANDS: Yes, In fact, this was done in the Zebra Doppler experiment. First, the effect of U^{238} absorption was represented by boron absorption, and scattering material was introduced in about the proportion which exists in a power reactor.

H. H. HUMMEL—*Paper 4A/4*: I am not sure exactly what Mr. Rowlands meant about the shift of the spectrum. Does he mean that there was a big concentration of fissile material in the vicinity of the sample which hardens the spectrum? It seems to me that if you worry about that, in the first place the small sample might have a mixture of U^{238} and plutonium, and compared with pure 238 this will be a fairly reliable measure of what will be encountered in a power reactor. I have not noticed any big discrepancy between what is inferred from this sort of measurement and a measurement of pure plutonium. I do not feel that the errors mentioned, with the small sample measurement, are very important. It is true that the hot-cold interaction effect has been a problem and there have been measurements which eliminated certain things to see what it meant, and there has been no difference. The cold neutrons can be shielded off and the signal altered at least. As long as not too small a sample is used— $\frac{1}{2}$ in. is a large sample—small sample measurements are good enough for an isolated rod. An overall hardening of the spectrum is imposed because of the change in fission and capture cross-sections. This is a 20% effect and is cause for some concern, but there are other uncertainties in the heated zone.

F. STORRER (Euratom; Centre d'Energie Nucléaire de Cadarache): A fairly large zone should, in principle, be better. The composition of the large zone which is heated should be the same as that of the driver zone around it. If the composition is changed in order to measure different components of the Doppler effect, flexibility is lost and the reactor must be reloaded every time. The advantage of a small sample is essentially that the composition can be pure plutonium. It does not make any difference. The theory shows that the sample composition can be arbitrarily changed—it does not have to be the same.

On the other hand, if the sample is too small the measurement of reactivity is insensitive. Probably the samples used in measurements are larger than they should be for the theory to be perfectly rated, in which case there is trouble in the interpretation. If you really want to use very small samples so that you can put in pure plutonium or U^{238} , another technique should be considered.

K. H. KÜSTERS (Euratom; Kernforschungszentrum, Karlsruhe): With regard to the Doppler coefficient, I wonder whether the temperature of a

sample would be below 600°C in critical assemblies? Do the Doppler coefficient experts feel that analysis of solid state properties below 600°C is necessary for Doppler coefficient calculations?

CHAIRMAN: Do you have a comment, Dr. Hummel, concerning the temperature of the sample in a critical experiment, as to whether solid state effects are sufficiently understood?

H. H. HUMMEL: Work has been done at Argonne on this, and some was done previously. There are some effects which show up, but they are not too large. It is easy to get above the region in experiments. This is something which should be taken into account.

J. W. WEALE (UKAEA): The discussion has centred on conventional measurements of Doppler effect by means of small or large samples in reactors. It might be useful to describe some measurements of the activation type, done recently by Dr. Perkin at Aldermaston. A small sample was placed at the centre of a spherical annular piece of beryllium. At the centre of the system was placed either a U^{235} or a Pu^{239} sample, and the fission rate was measured at temperature. U^{238} samples were also placed there. This experiment gave some reasonably direct results. These showed agreement between calculation and experiment for U^{238} , but the U^{235} and Pu^{239} effects, although positive, were about one-sixth of the predicted value. These results were published by Dr. Perkin at a recent Washington conference.*

R. D. SMITH: Professor Yiftah showed dramatically the effect of different plutonium isotope compositions on Doppler effect. I would question his choice of plutonium A, B and C. These compositions appeared many years ago, possibly in a paper by the Chairman. I wonder whether plutonium C, in particular, is a realistic composition, whether these calculations should not be done with a composition corresponding to the equilibrium isotopic composition eventually reached in a fast reactor cycle.

S. YIFTAH—*Paper 4A/2*: I think you are right. We have done this. The results that I showed were only for these three compositions because of the need to relate them to previous calculations of the static parameters and also the effect of the isotopic composition on the critical masses.

CHAIRMAN: I wonder if any of the groups who are measuring Doppler effect or who plan to do so would speculate on the number of such measurements over the world that they feel is needed to be adequate. Is this number 10 or 100, and so on?

* Paper F8 Washington Conference on Neutron Cross Section Technology 22nd–24th March 1966. J. L. PERKIN, UKAEA.

K. WIRTZ (Germany): Probably in a facility like SNEAK about three or four experiments a year can be done. We know the number of facilities, and thus we can work out the number of experiments per year which is possible. I imagine that half of them will be identical. Perhaps it would be possible to suggest what the best experiment would be.

F. STORRER: What is needed is not so much a large number of experiments as good ones. When reading the literature, I am struck by the fact that the experimental procedure is always presented in very great detail, including all corrections, but in those papers very little is usually said about the theoretical interpretation. I think that if we want to benefit from experiments performed in the United States or in Great Britain or anywhere else, it is essential to have a good description of how the experiments were interpreted. I think that too much emphasis is put on the description of the experimental aspect but not enough on the theoretical aspect.

F. M. HECK (Westinghouse): I would also say that in interpreting the results, we have difficulty in understanding the experiments and the descriptions which are given.

J. ADAMSON (UKAEA): I wonder whether any of the speakers could say what principles should be embodied in a good experiment to measure sodium worth? The sodium in a fast power reactor is a fine balance between leakage effects and the capture effects and the spectrum change. Suppose that a particular power reactor of specific design is envisaged and some idea of the worth of the sodium in the reactor is required. There is a zero energy reactor on which you can try to do the experiment. In that reactor you may not have exactly the right materials to mockup the power reactor. How difficult is it to simulate the adjoint function and the normal flux spectrum at the same time?

The plates or cylinders or slugs in the zero energy reactor will have a certain thickness which is different in all probability from the thickness of the fuel pieces in the power reactor. How do you infer from the results of the zero energy reactor what the sodium effect would be in the power reactor, with its own particular heterogeneity?

R. AVERY—*Paper 4A/1*: In reply to the second question we experimentally observed quite large differences, and one-dimensional transport calculations were made—admittedly, so far they have not been very good—which seemed to give not too bad agreement with the changes that we observe. Probably two-dimensional calculations would be required. I would hope that doing such calculations of the experimental configurations and also (something that we are planning, and perhaps others are) fabricating special drawers

which might have much less heterogeneity, one might be able to confirm this point.

The first question pointed to the fact that one is interested in a sodium void coefficient under a very large number of possible configurations, perhaps something which one would not be able to get a mockup of in the critical experiment. I have in mind the effect of higher isotopes, and so on, which one might not have available. I suppose that the only answer would be to take the measurements under a sufficiently wide variation of compositions so that one might calculate it.

However difficult these problems are, the basic issue, I think, will still be determining the sodium void coefficient for particular specified configurations which might be done by taking more care and getting proper samples. There is also the determination of the question as to what are the actual void distributions which might occur under various conditions and what their significance is, if any, in relation to reactor design. But that is a question which someone else might answer.

J. ADAMSON: On the previous core of Zebra—core 6—we deliberately set out to try to make a measurement of the sodium worth which might be of some use in the design of the Prototype Fast Reactor. But we did not have exactly the correct material to mockup the prototype. We were unable to simulate both the normal flux spectrum and the adjoint at the same time. We got a very good match for the normal flux spectrum, but could not at the same time get a good match for the importance spectrum.

R. AVERY: But, then, one is left with not being able to interpret the answer directly as it might apply to a prototype. But if one takes the measurements under a variety of conditions one might have something to go on.

F. STORRER: I think that in most cases it is a mistake to try to measure in one integral experiment directly the parameters pertaining to a power reactor. It is good in some cases, as Dr. Smith pointed out. They wanted to find out whether the Doppler coefficient for the prototype reactor was negative. However, in most cases it is impossible to measure directly a quantity like critical mass or sodium coefficient. It is essential to make clean experiments which can be interpreted and methods for interpreting these experiments properly and for applying the results to power reactors have to be developed.

N. V. KRASNOYAROV—*Paper 4A/3*: I agree with Dr. Storrer that experiments must be carried out under conditions which give clean results so that they can be compared satisfactorily with the calculations.

There are other interesting experiments which are nearer to the prototype reactors and are amenable to direct experiments. One can almost never get the same results on the mockup that one can obtain by direct measurement

on the prototype. One should aim at making experiments so that one can transfer the result with a minimum of correction.

In regard to the experiment on the Doppler effect mentioned by Dr. Smith, the Doppler effect was measured in a little zone, and some recalculation is required in order to apply the result to the power reactor. Similarly, exact data are needed for sodium void coefficients. One has to measure the effect of sodium removal, and the spectrum at that time has to coincide with the correct power reactor spectrum. For that reason, one has to have a large enough volume, and, naturally, the distribution of worths will be different, and that can be recalculated in the same way as with the Doppler coefficient.

CHAIRMAN: I wonder whether anyone in the audience can define the minimum quantity of material or the minimum volume of material that will be required to get a good measurement of the sodium void coefficient? Dr. Storrer, can you define "a good experiment"?

F. STORRER: The sodium coefficient is a very difficult question, but one should not try to measure in one experiment the sodium coefficient while simulating exactly the heterogeneity of the power reactor, the fission product content of the power reactor, etc. Separate experiments should be performed to investigate all these aspects.

Is it usually necessary to measure the isothermal Doppler coefficient, while what is important in a reactor may be the power coefficient or something else which is much more complicated? I do not think that one can ever measure directly the quantity of most interest.

CHAIRMAN: Let me be more specific. Suppose one is interested in learning in an experiment the contribution of plutonium 240 to the sodium void coefficient and one wishes to ascertain how much plutonium 240 one should keep fabricated in large form in order to get a sufficiently accurate experiment in order to give the theorists something to do the best they can on. Have you considered this?

F. STORRER: No, we have not. However, I would say that first you should try to see whether you have an adequate method of calculating and measuring the sodium coefficient, for instance, in the absence of plutonium 240. Then you should carry out other experiments to see whether you can calculate correctly, for instance, the worth of plutonium 240 which is related to the cross section of plutonium 240. All the answers cannot be obtained by logical deduction, but by carrying out several experiments verification may be obtained. If you understand something well, then you can eventually isolate a parameter which is not well known. For instance, if a Pu^{240} experiment is not successful—if the calculations and the experimental results do not coincide—you may discover that the Pu^{240} cross-section is not correct, and so you may put more effort into that. Is there another method of measuring the sodium

coefficient than by building a reactor and running it to 50,000 MWd/t burnup and taking the sodium out?

CHAIRMAN: I agree in principle about trying to understand each part. I think that there is also merit in trying to measure the integrated effect if one can. The question is: What expense is necessary, and is this expense warranted?

R. D. SMITH: Since we do not understand the sodium coefficient completely, if at present one wants to make a measurement in the zero power reactor one has to have all the fuel and all the sodium. But it is a very expensive opera-

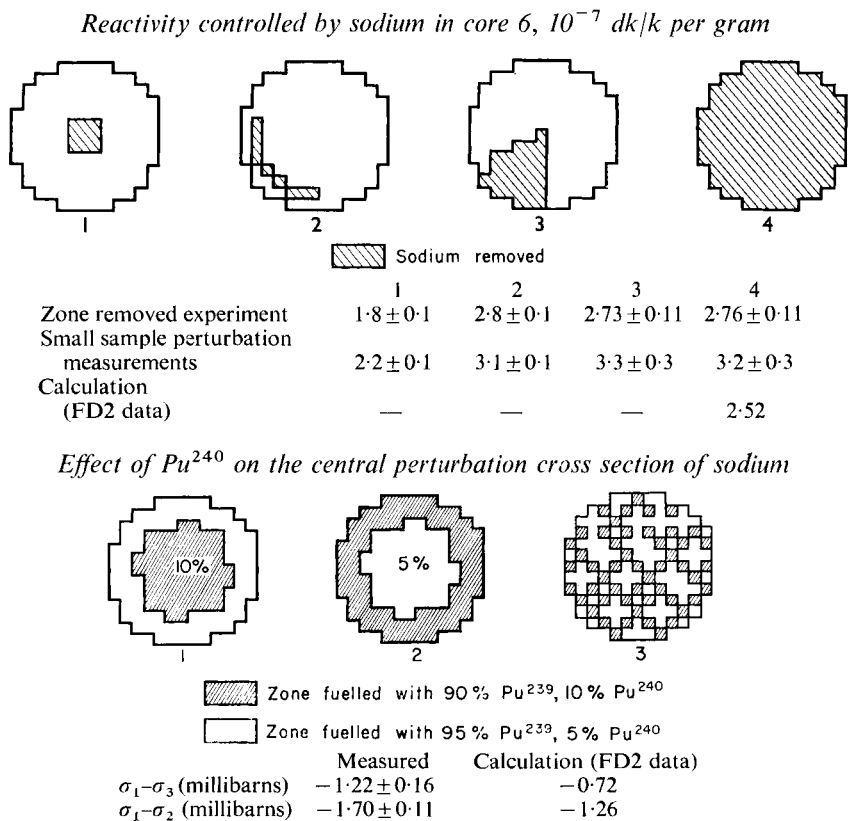


FIG. D1

tion if there are sets of fuel with different Pu^{240} contents. In Zebra there are batches of fuel with different Pu^{240} contents, and we have tried to find some effect of the Pu^{240} on the sodium coefficient. We have also done tests to obtain an indication of the sodium coefficient, and you get that measurement on a small sample.

The different zones of sodium are given at the top of Fig. D1. We removed

zones from different parts of the reactor and measured the perturbation effects. There is one calculation to compare this with. All the numbers are different, but not very different. They give some hope that this sort of thing gives a good measure of what you get in a complete power reactor, bearing in mind all the other questions of heterogeneity and plates and so on which remain unanswered.

With the Pu^{240} we had a 10% and a 5% zone and a third experiment in which the 10% and 5% elements were mixed. By taking differences from one to the other one can have some idea of the effect of Pu^{240} and compare it with the calculations in this situation, using the FD2 data in Fig. D1. These are recent measurements. At present I would not care to put a stronger interpretation on these than to say that our calculation of the effect of the Pu^{240} on the sodium void coefficient does not seem to be too far wrong, but I would not say that it has also proved that it is right.

J. W. WEALE: I should like to ask Dr. Avery a question about the fission neutron spectrum. He suggested that there may be more neutrons in the low energy region than is at present thought. Can he say whether his new figures are sufficient to account for the discrepancy of about 10% between the calculated and the experimental lifetime to which he referred in Dr. Hummel's paper? Does he think that this is a more likely explanation of the discrepancies than errors in the scattering calculations or misinterpretation of experimental results?

R. AVERY: It is the result of numerical studies, and it is not anywhere good enough to explain the discrepancies with respect to the neutron lifetime. In systems which are fairly well moderated, either oxide or graphite, there is very little difference in the spectrum. However, in models of full scale experiments with sodium, it has a fairly significant effect on the very low energy spectrum.

Below about 50 keV, the measurements which were obtained by Dr. Meadows at Argonne indicated the possible increase in the spectrum. It starts at around 50 keV and can go up by a factor of 2 at much lower energies. But the error bars are quite large, and, as a matter of fact, they overlap, just barely missing the theoretical estimates. It is really only an indication that there is a measure at very low energies. It is not large enough to have any significant effect on a large number of the characteristic discrepancies which have existed with respect to neutron lifetime or boron work.

J. W. WEALE: Dr. Avery mentioned the effect of moderation. It seems that the possible misinterpretation of experimental results is more likely to result in these cores because there is a time dependent effect in the spectrum. You have a fair amount of slowing-down material there.

The other important point is the possibility of misinterpretation of the experiment. One may have taken an incorrect value of α . This might be very important in explaining the current discrepancies.

C. P. ZALESKI (France): With regard to the question raised by Dr. Avery, for power reactors as envisaged at the present time the contribution of the sodium coefficient is of the order of magnitude of 10% of the part of the power coefficient which is significant in safety work. With regard to the accidental situation, these situations are so little known—the phenomena are so important but are not known—and the lack of precision in our information is such that we do not need accuracy in respect of the effect of the sodium coefficient. I do not think that you can get higher accuracy of the total effect.

H. H. HUMMEL: Concerning low energy spectra, the discrepancy in the Doppler effect of fertile material is the most convincing thing I have seen. Some of the other things are rather uncertain. In regard to neutron lifetime and the boron work, there are a lot of ifs, but the Doppler effect seems straightforward. It seems difficult to blame it on anything other than the low energy spectrum.

WEALE: Dr. Smith did not mention in his paper that there have been some spectrum measurements made on one of the Zebra cores in which the Doppler effect was measured. There is some discrepancy in the lower energy regions, but it is not sufficient to explain the differences in relation to the Pu^{239} or U^{235} effect. The results that we have indicate that the spectrum in the region below about 10 keV is about 30% too small as predicted compared with the experiment.

K. H. KÜSTERS: In regard to heterogeneity calculations, what is mostly included is the heterogeneity of the drawers. How much does this affect the result? There are two heterogeneity effects—the fuel plates and the drawers. I wonder whether there has been any investigation in this field.

H. H. HUMMEL: We have not looked at this. I do not see that it can be very important. There are two different types of heterogeneity effects. One is concerned with the fine structure flux, with the changes between the fertile and fissile materials and other resonance effects at low energies. Work done at Argonne has been concerned primarily with the high energy end. We are beginning to look at the lower energy end, which will be important, but we have not done much on it yet. It is, I think, tractable. We are now going to alloy the fertile and fissile materials. So this reduces somewhat the magnitude of the problems, because we do not have to worry about these effects. Certainly with very dilute reactors you have to be careful about the inter-

pretation of the effect of the fission and capture cross-sections, taking into account resonance shielding.

K. H. KÜSTERS: I wonder whether it is possible to interpret the experiments with sodium void effects and Doppler effects for the purpose of large power reactors (where there is less of a heterogeneity problem).

H. H. HUMMEL: With regard to the sodium void effects and the spectrum component, we feel that if we get reliable central Doppler coefficients we can use these to reduce the uncertainties of the cross-sections to a manageable degree. I feel that a combination of this and a limited number of experiments with a U^{235} fuel core would be sufficient to pin down the leakage component pretty well. Between those two, I would feel fairly happy about it.

CHAIRMAN: While we are on the subject of heterogeneity, I notice that on page 461 of Paper 4A/3 reference is made to a correction for heterogeneity of 0.1% for k -effective. While I have not looked at the detail of this assembly, thinking back to some of the earlier ZPR-3 assemblies which used plates of U^{235} which were certainly no thicker than these, I believe that we sometimes calculated larger corrections which were due to heterogeneities. I wonder whether the Russian authors could comment on this number and say whether they feel that the corrections for the BFS criticals are too low in this range or whether there is a range of experiments where the corrections for heterogeneities using such plates are larger?

N. V. KRASNOYAROV: When we started to calculate the effect of heterogeneity, it appeared that the effect of heterogeneity in different assemblies was not the same. It must be pointed out that the assembly was not ideal from the point of view of the size of the pieces, which were not the same as in a power reactor. So we had to keep to the average concentration of fuel. In other assemblies with regular cell patterns the same calculations were made which showed that heterogeneity had a large effect on the breeding ratio, and here the agreement with experiment was less good.

But the errors were not too much bigger. In our experiments when we got other heterogeneities with doubled plates we got different results.

CHAIRMAN: Do you feel that one can use enriched uranium building blocks, or plutonium blocks equivalent in thickness, to study dilute fast power reactors, or does one have to go to a mixture of fissile and fertile material in the building blocks in order to keep the heterogeneity correction to a reasonable value?

N. V. KRASNOYAROV: For the investigation of diluted systems, it is very desirable to have a mixture of fuel which would not have such heterogeneity, but, unfortunately, such a type of pellet would not allow the making of a flexible assembly for different mixtures; and that is why we get systems with

heterogeneity complexities that have to be accounted for, and that is why we have to put in corrections. If we were in a position to calculate all the effects and results in such assemblies, then we should be able to calculate them for big reactors where the heterogeneity will be less, but for that purpose we have to be sure that in taking into account the corrections for different mixtures we are not making errors which are overlapping. That is why in heterogeneity experiments one has to investigate in great detail the effects so that everything is in agreement, and not only the overall results.

CHAIRMAN: Dr. Smith's Table 5 showed the comparison between prediction of criticality and experiments mostly better than 1%—slightly worse than 1%—on one experiment. My impression of the knowledge of cross-section information is that if one started such calculations having heard nothing about critical experiments in the last 5 years, one might find oneself in the position that Dr. Hummel described previously concerning the choice of parameters. One might choose parameters which fit the critical experiment and one might not. To what extent does one have a happy situation in agreeing with criticality, and to what extent is this empirically obtained?

R. D. SMITH: These calculations using the FD2 set are made with data which are direct and not adjusted in any way; and which anybody might have deduced and calculated without ever having such critical experiments, but that would be luck, because this is known to contain cancelling errors. We know that the U^{238} data are wrong, we know that the Pu^{239} data are wrong and we know that the U^{235} data are wrong. So these answers come right as a result of cancelling errors. On the other hand, it seems that possibly on average the data are perhaps a little better than they might be from some of the known scatter in the microscopic measurements.

J. B. NIMS: Have you calculated the SEFOR reactor?

R. D. SMITH: No.

K. H. KÜSTERS: In Karlsruhe we have started an investigation for separating the effect of several of these uncertain cross-sections on k -effective, and later we shall compare all the cross-section data together,

In answering the question by Dr. Nims, I would say that we have calculated at Karlsruhe the k -effective of the SEFOR mockup for our own and for the Russian set, and we got an agreement without any heterogeneity correction of 4% in the critical mass. We got 325 kg of plutonium with our Karlsruhe set, and 360 was obtained on this assembly. It is known that heterogeneity reduces the critical mass a little and it goes in the right direction.

CHAIRMAN: Was it the same number with both cross-section sets?

K. H. KÜSTERS: No. The Russian set for SEFOR gave better results for

critical mass. I suppose that it is $\frac{1}{2}\%$ or 1% out in the critical mass. I do not remember the figure now. However, I would say that it is quite lucky.

I should like to ask Dr. Avery or Dr. Smith whether there have been recent measurements of the inelastic cross-section of Pu^{239} or the higher plutonium isotopes, because these are very uncertain data.

R. AVERY: Not to my knowledge.

J. L. ROWLANDS: I believe that measurements are in progress at Harwell, but the results are not yet available. However, we believe that the uncertainty in the U^{238} inelastic scatter cross-section is much greater than that in the Pu^{239} .

R. D. SMITH: I should like to ask Dr. Avery and Dr. Krasnoyarov a question about the breeding ratio. I should like to know with what accuracy they think they can calculate the breeding ratio on a large fast reactor and measure it in a zero power reactor.

R. AVERY: About a year ago we were asked to make an estimate of this, and, as far as I know, the situation has not changed materially. I imagine that then the error was of the order of 0.1 or 0.15 in BR in the systems of interest as opposed to the systems that have been studied. In the critical experiment one does better since one has at least some of the relevant quantities which have been measured.

N. V. KRASNOYAROV: We have not really tried to calculate the accuracy of our breeding ratio calculations. We base it on cross-sections, and taking into account the cross-sections, we get some results. The accuracy of the experiment was calculated. This used the errors of measurement of the spatial distribution of the capture and fission rates. These errors were given in our report. I have already mentioned today an error bigger than that given in the report. We are very seriously interested in this question, and our experiments at the moment do not show any grounds for increasing it further.

J. H. BOWEN (UKAEA): I have a question about SEFOR. We are told that we may expect direct measurements of the Doppler effect. I wonder to what extent this will be obscured by other prompt coefficients. What magnitude of correction must be made to the SEFOR measurements before the Doppler effect is measurable?

R. AVERY: As I recall the expansion effects, they are minimized by some segmenting of the fuel, but I do not recall whether the present pin is of two or three segments. My answer must be based on my greater recollection of our plans on FARET, which is now no longer on the experimental programme. The calculations showed that the other effects, such as expansion of fuel and other things, were in the range of 20–30% of the Doppler coefficient. I believe that on SEFOR more work is being done to minimize the expansion. I would

expect that it would be somewhat better. This attributes corrections to those items that one can identify. There is always the risk that there may be some spurious reactivity coefficient. I wonder whether any of the people from Karlsruhe might be somewhat more familiar with this point.

K. WIRTZ: As far as I understood the question, we know that if the fuel is segmented and has the dimensions that have been mentioned previously—thick rods, etc—then the response of the reactor is a fast power coefficient which might contain, besides the Doppler effect, a bit of the fuel expansion. But we feel that all this is a very fast effect that will also appear in the final reactor. We believe that we can get the exact power coefficient.

J. H. BOWEN: I can understand, for normal purposes, that the two coefficients may be mixed together, but for special safety purposes one is interested specifically in the Doppler coefficient. Could you add what the magnitude of the two effect is? Would it be two-thirds, one-third or 50%?

K. WIRTZ: I think that it would certainly be more than 50%. That is a personal guess based on what I know of the construction. Perhaps it might even be more than 60%.

R. AVERY: My guess would be that 20% other than Doppler would perhaps be high.

CHAIRMAN: I agree. I believe that there will be other uncertainties, such as fuel temperature. It is not as if one knows the fuel temperatures precisely, and as 80% of the reactivity effects can contribute to the Doppler effect, it still is a reasonably complex measurement.

H. H. HUMMEL: This was part of the original argument for the SEFOR reactor. The argument that you cannot measure fuel temperature but that you mockup what it is by some means, so that the temperature of the fuel pin in SEFOR will be the same as in the power reactor and you are actually measuring two different things at the same time—the thermal conductivity of the fuel and the Doppler effect. The argument was that this was desirable. I do not know whether the feeling of General Electric has changed on this in the meantime, but I know that it was felt to be a bonus for the experiment originally.

CHAIRMAN: It might also be a necessity.

C. P. ZALESKI: I feel that there may be a difference in the conductivity of the fuel irradiated and non-irradiated, and I also think that the fuel will not be greatly irradiated. It is not obvious whether we are really measuring the right coefficient. Unless the whole of the fuel is irradiated, all the values relating to the conductivity of the fuel can vary. If the power coefficient is to be representative of the power reactor, then the thermal properties of the

fuel must be the same for unirradiated fuel and irradiated fuel, and I do not think one can be sure about that now. I do not know whether there are any plans to make experiments in SEFOR with unirradiated and irradiated fuel to many thousands of megawatt days per ton.

CHAIRMAN: I believe that the power reactor is small for the core volume and that it would take a very long time to get the burnup you refer to.

C. P. ZALESKI: The conductivity of the fuel can change.

H. H. HUMMEL: I agree with that point. The discussion a little while ago was pointing out that you would, after all, be measuring a fuel temperature coefficient which might be the same as for a power reactor, but I think that this is not true, because the expansion effect will not be the same in SEFOR as in a power reactor. I believe that the expansion should be small. Since they have gone to the trouble of segmenting the fuel—giving a 20% effect or less—this is probably not a major concern in the experiment.

N. V. KRASNOYAROV: With regard to the accuracy of our Doppler effect and the accuracy of the temperature coefficient, there are inaccuracies in making a comparison with experiments on the reactor which are connected with the fact that in the reactor you will have a heterogeneity different from the heterogeneity of large reactors. In the case of large reactors we can be negligent about the heterogeneity, but here it will be important. Besides that, if you neglect the heat conductivity, then the temperature of the fuel is always dependent on the size of the coolant space in the assembly, and that will also have its effect on the temperature of irradiation. If you consider the accuracy of the Doppler effect, then you will find that high accuracy will be very expensive.

F. M. HECK: Dr. Avery previously described the programme of TREAT tests for fuel meltdown and then the analysis of the fuel melting in cores (session 3). There was nothing said about any need for doing fuel core experiments of the SPURT or BORAX type. Was this an omission because he feels that it is not necessary or because it is not yet included in the programme?

R. AVERY: This question first arose at least 10 years ago. It is not in the programme now. I think that there is some thought in the U.S.A. that eventually some kind of more integral type of experiment will have to be conducted. This may take many different forms. Perhaps this is a good example of changing thoughts. For example, had this been pursued 5 years ago, a meltdown phenomenon would clearly have been investigated, and it would probably not have taken any great consideration of the sodium expulsion and related effect. I think that if one were to do such an experiment now—it is a completely undefined thing—one would be much more concerned with questions of sodium voiding and possibly propagation effects of

the sodium voidage effect. I also think that if one were to do such an experiment now one would also attach great importance to the fact that the irradiated fuel might behave quite differently from the unirradiated fuel.

There is thinking along these lines but it is not part of the programme now. Dr. Nims and APDA some few years ago studied a meltdown type facility. I should not be surprised if some such more integral destructive or BORAX type experiment were done at some time or other.

J. B. NIMS: I should prefer to see some specialized measurements made: some experiments with meltdowns in floating sodium to see how the fuel disperses, rather than a complete meltdown.

F. STORRER: An experiment similar to the one suggested is being studied in Cadarache. A decision has not been made to perform it, but it is being actively studied, and it will probably be performed. It would be an experiment in CABRI, which is the French analogue to SPURT. It is a water cooled reactor. However, one would be thinking of putting into the centre of CABRI a fast zone composed of Rhapsodie fuel assemblies. One would investigate the propagation of meltdown from one subassembly to the next. One would examine a subassembly and see whether melting was propagating to the neighbouring ones. This is somewhat cheaper than blowing up the whole reactor.

K. WIRTZ: I think that an experiment such as Dr. Avery suggested would probably only be done after the results of SEFOR had been achieved. One can then learn how the power excursions really go. Or does Dr. Avery think that it would be quite independent of the SEFOR experiments?

R. AVERY: Generally, I think that one would wish to perform any integral measurement only after one has understood as much as possible of the phenomenon. The major objection to such experiments is, of course, that you get one point but you do not know how to interpret it. My reaction is that there would not be too much interaction with SEFOR. At least, I would not think that the demonstrated Doppler effect is so crucial on the uncertainty. I think that one would wait for it. It may well be that preparations for any such experiments, by their nature, would take one beyond the time when SEFOR gives its results.

H. H. HUMMEL: I agree with that. Undoubtedly, the uncertainties of the meltdown business are of a different order of magnitude from the uncertainties of the Doppler effect. The SEFOR experiment is designed to remove any doubts about the Doppler effect in an operating power reactor, but with regard to the action of the Doppler effect on a meltdown, I feel that there are other things to worry about than how well you can calculate it.